

Interlinking Research Infrastructures: A Focus on Geoscience and Climate

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EPOS
EUROPEAN PLATE OBSERVING SYSTEM

**SUMMER
SCHOOL**

ESPAI CRÀTER, OLOT (SPAIN)
JUNE 23 - 27, 2025

The First EPOS Summer School is organised in the framework of the EPOS ON Project, funded in the Horizon Europe programme under GA n° 101131592

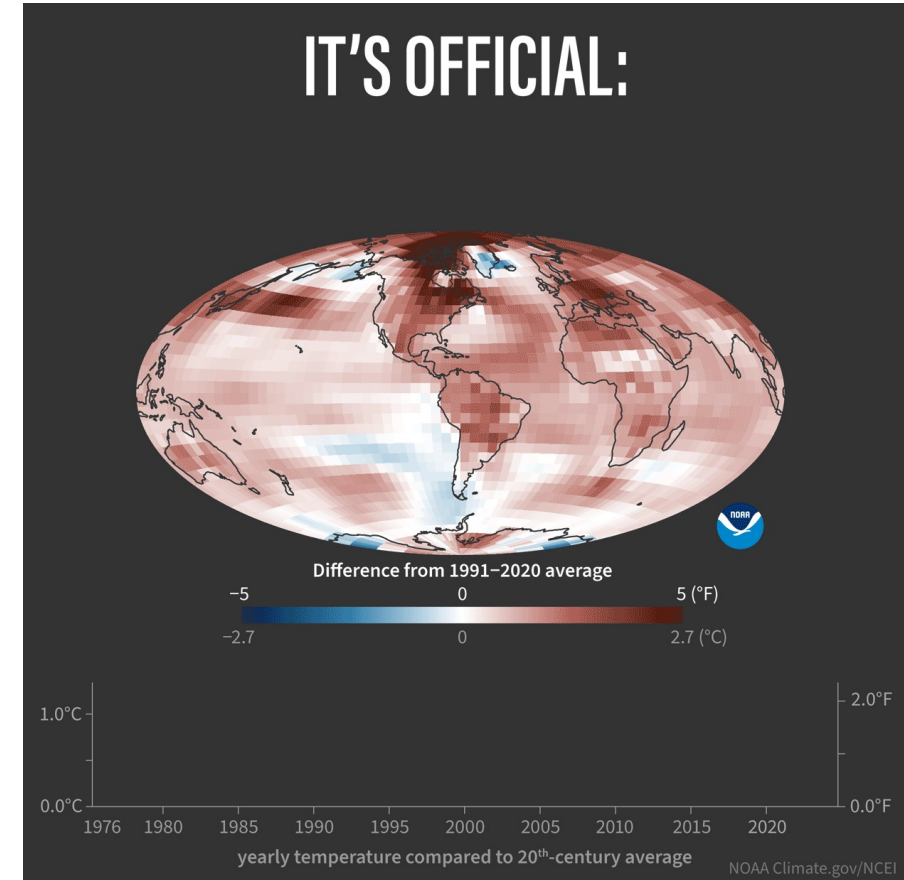
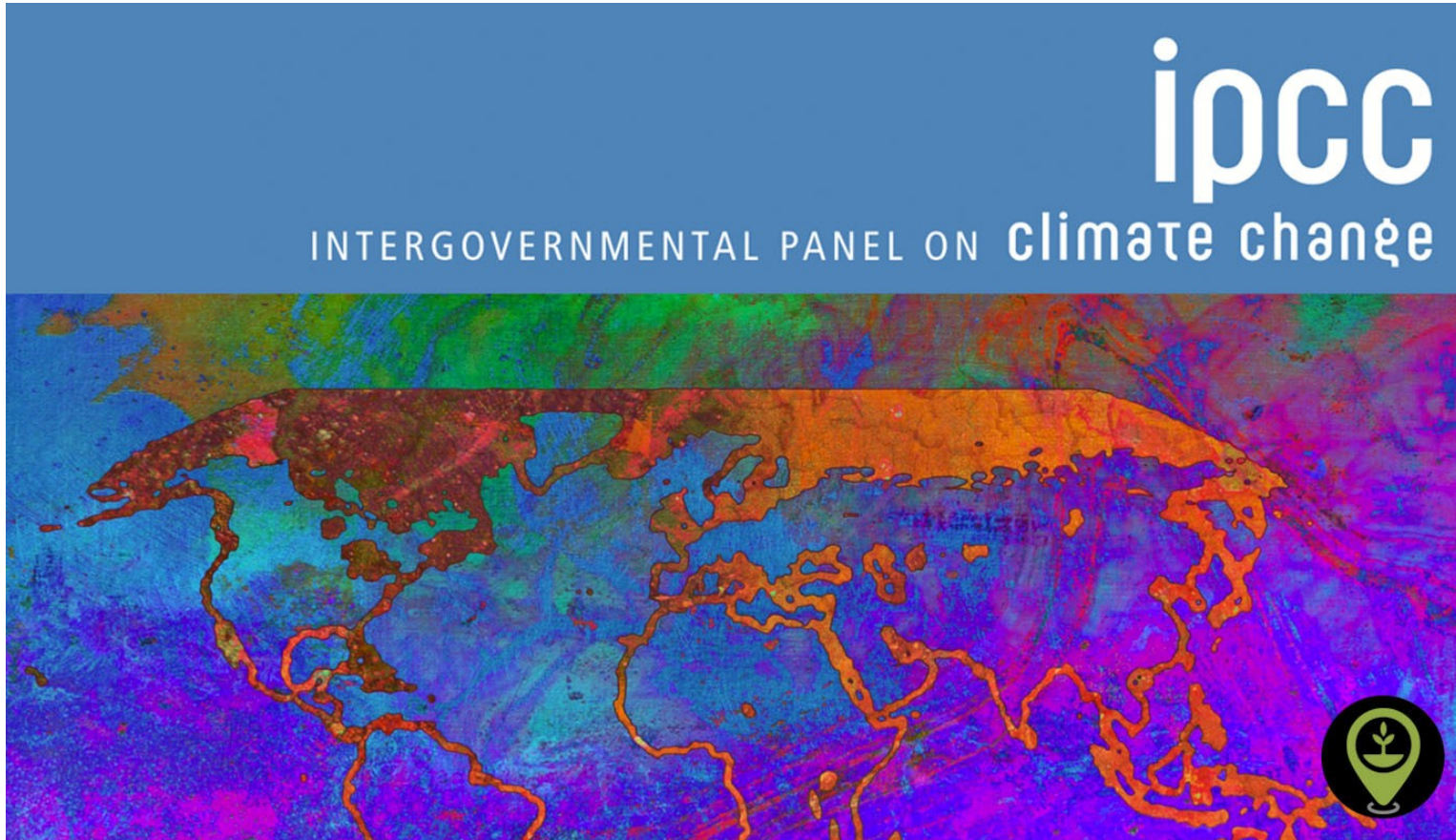
EPOS
EUROPEAN PLATE OBSERVING SYSTEM

 **Funded by
the European Union**

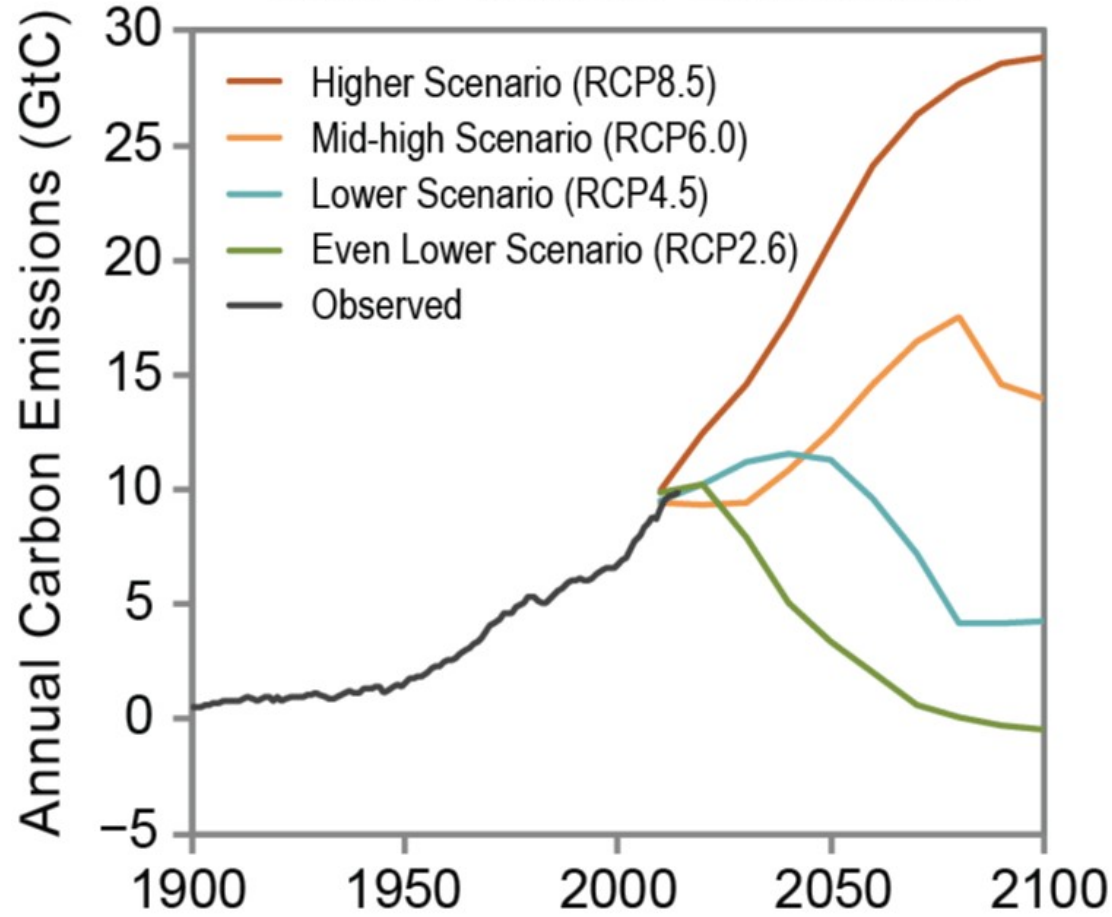
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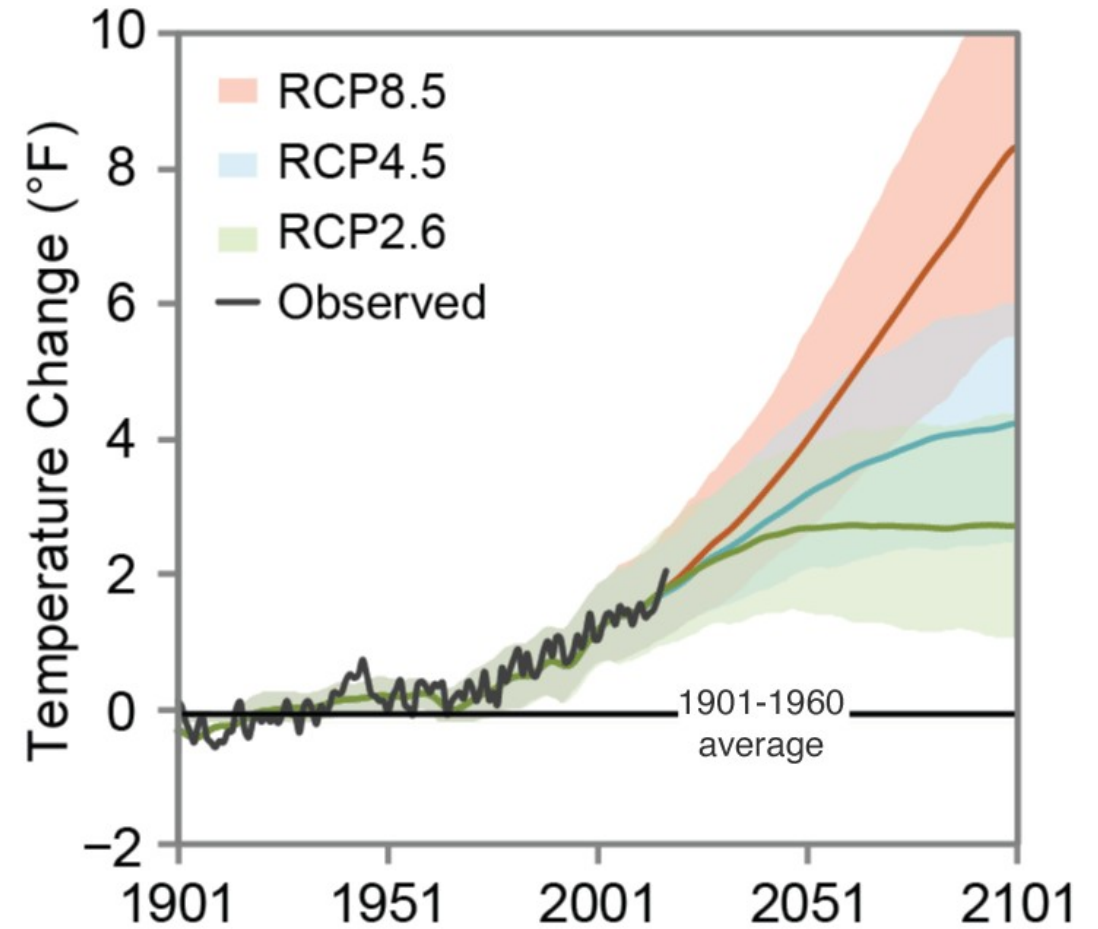
Climate change: from projection to observation



Projected Annual Global Carbon Emissions

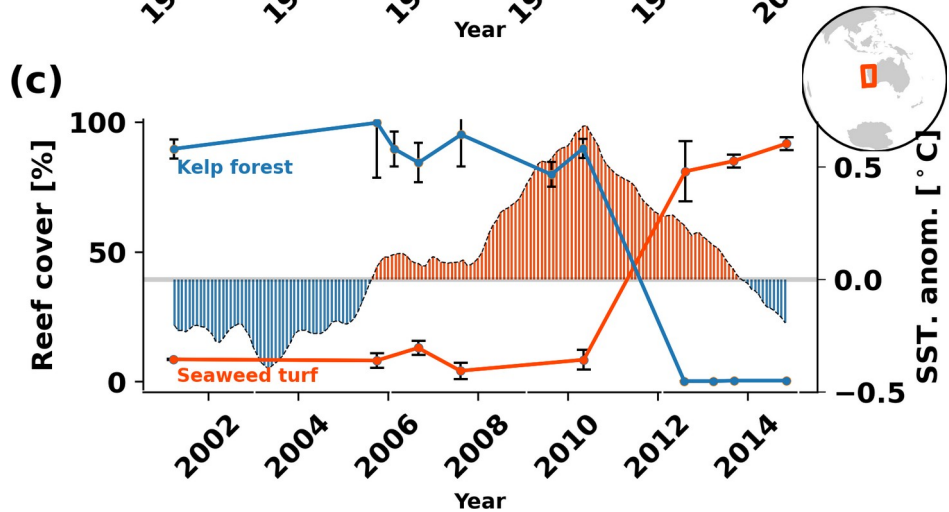
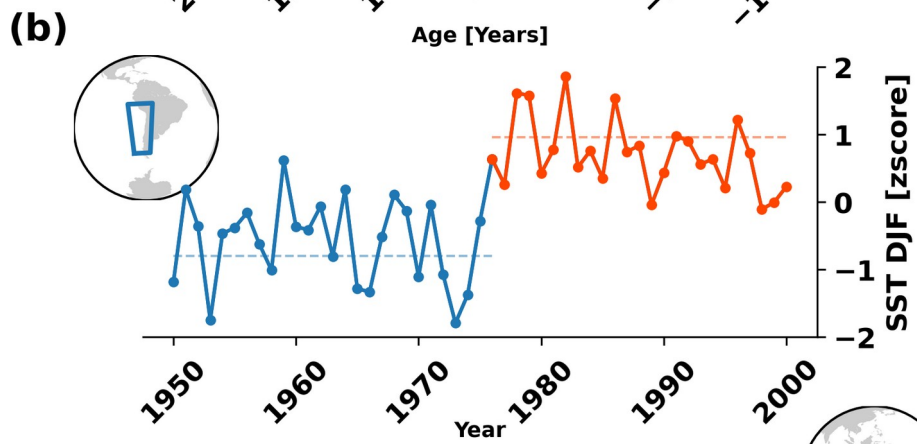
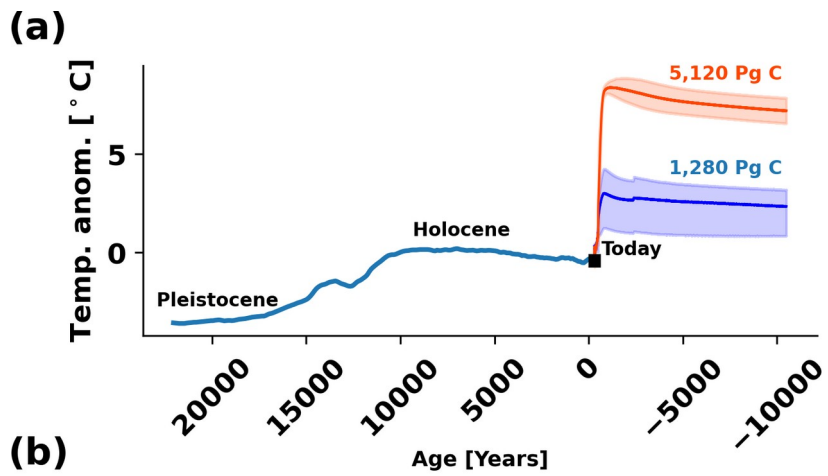


Projected Global Temperatures



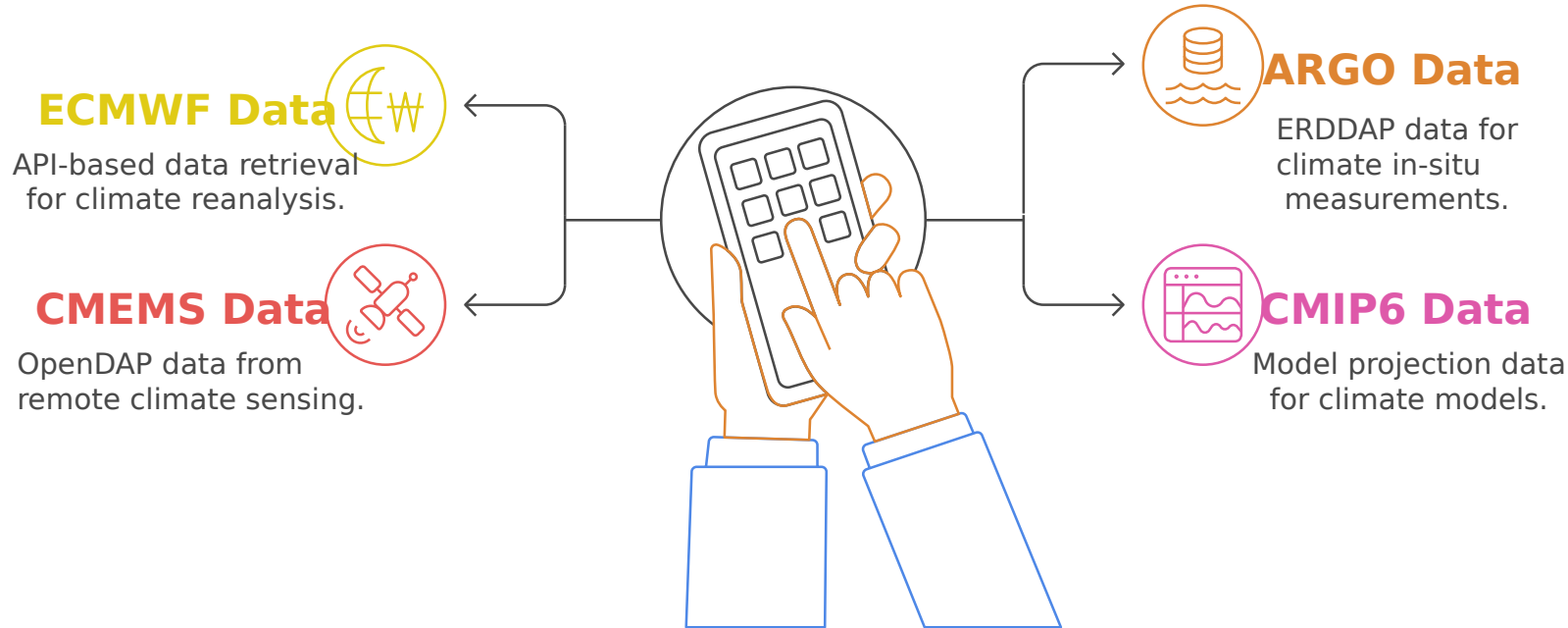
2017 Climate Science Special Report, Figure ES-3

(left) Hypothetical pathways of carbon emissions ("representative concentration pathways," or RCPs) throughout the twenty-first century based on different possible energy policies and economic growth patterns. (right) Projected temperature increase relative to the 1901-1960 average depending on which RCP we eventually follow. Image by Katharine Hayhoe, from the [2017 Climate Science Special Report](#).



Heinze C., Federike R., Ramanantsoa J., and 32 co-authors, 2025 (*Biogeochemistry Journal*)

Major Service Providers and Applications



- **Climate Reanalysis:** ECMWF (API-based data retrieval)
- **Climate In-situ:** ARGO (ERDDAP)
- **Climate Observation:** CMEMS (OpenDAP)
- **Climate Model:** CMIP6 (Model projection)

Application1: epos_argo_cmems.ipynb

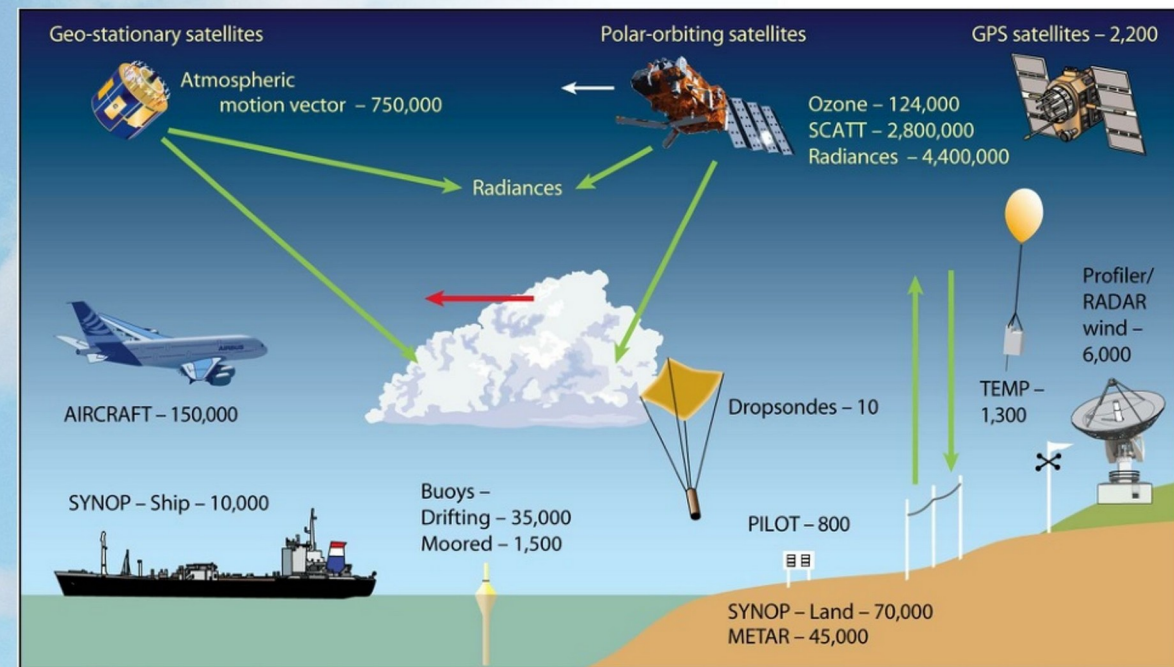
Application2: epos_ecmwf.ipynb

ECMWF

European Centre for Medium-Range Weather Forecasts

- ECMWF provides a wide range of meteorological and climate variables, especially through datasets like ERA5 and SEAS5.
- Largest data provider in ocean and atmosphere system

Dealing with large amount of multi-dimensional data



WIS Enablers, Geneva 2010 - 4

© ECMWF



**Science-based operational services
supporting environmental monitoring and
research**

ECMWF

European Centre for Medium-Range Weather Forecasts

- ECMWF:

<https://cds.climate.copernicus.eu>

- API:

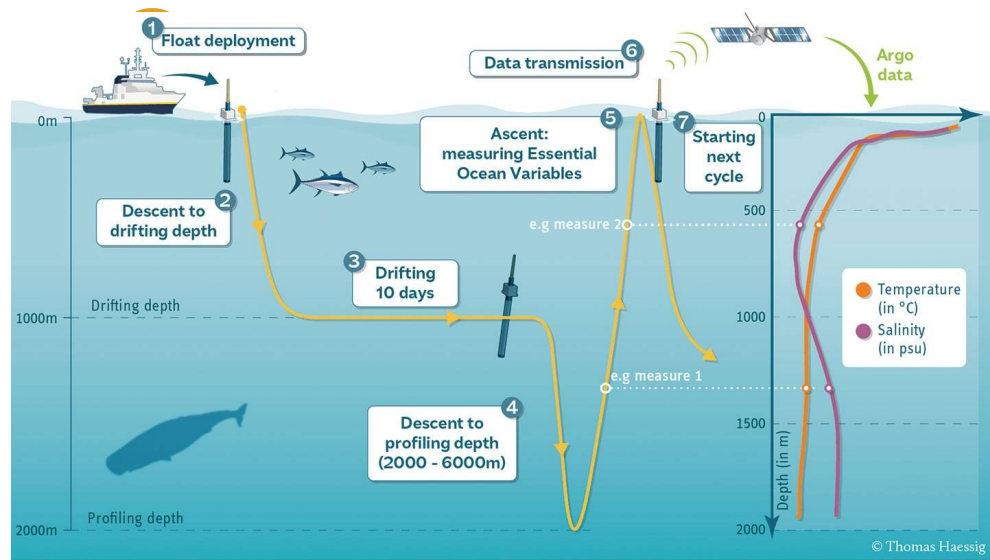
<https://cds.climate.copernicus.eu/how-to-api>

- How to install:

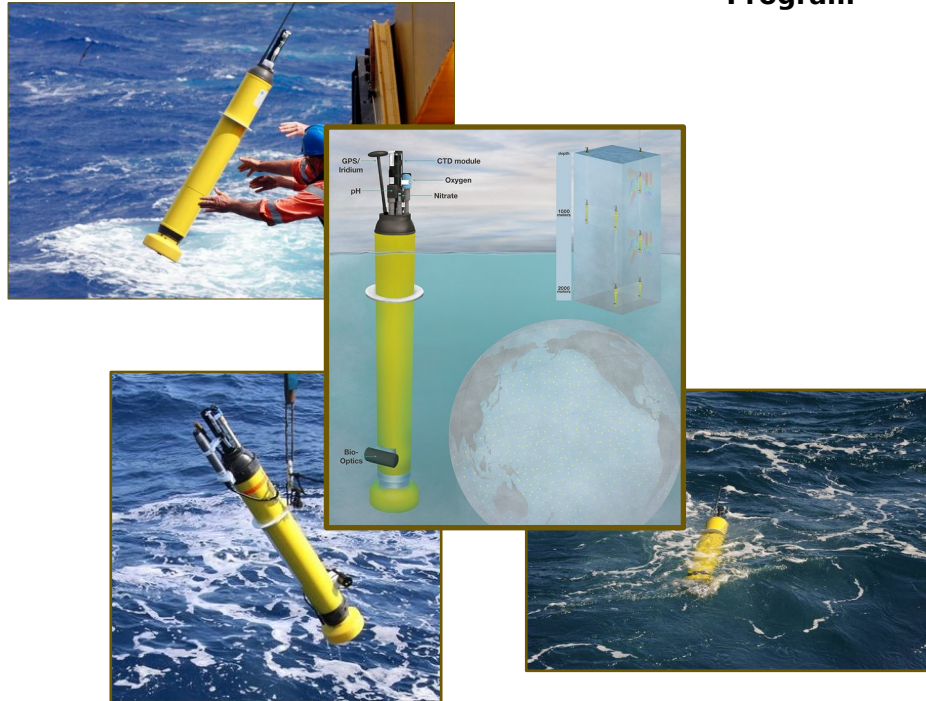
<https://confluence.ecmwf.int/display/CKB/How+to+install+and+use+CDS+API+on+macOS>

- Package dependencies:

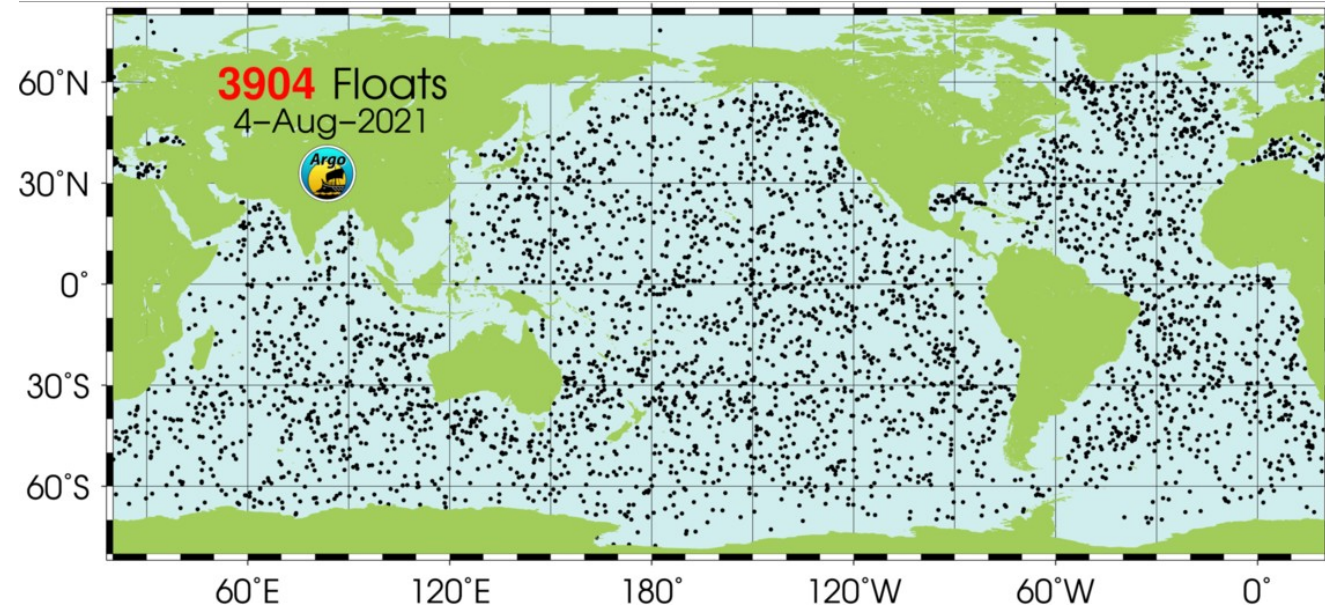
```
> import cdsapi
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Global ARGO Program

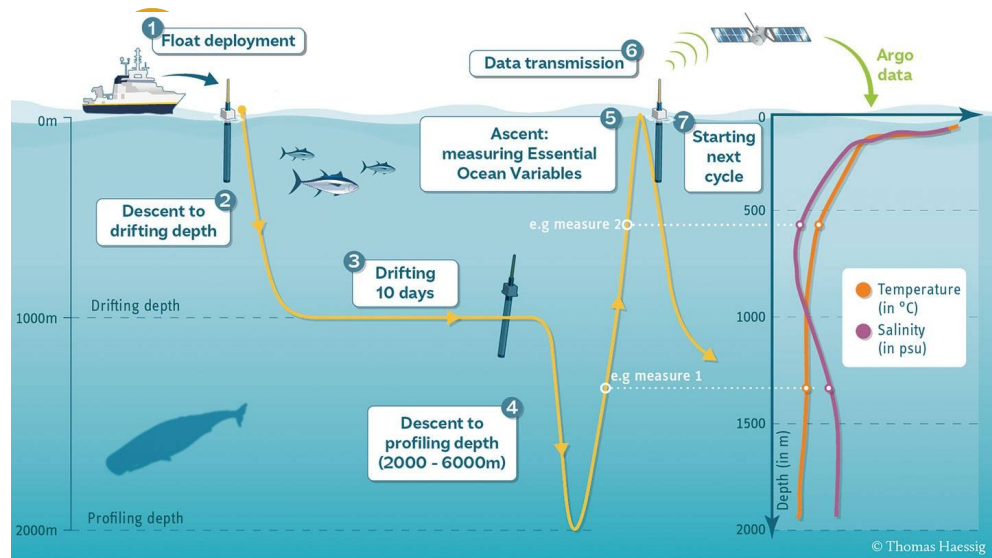


ARGO (BGC)

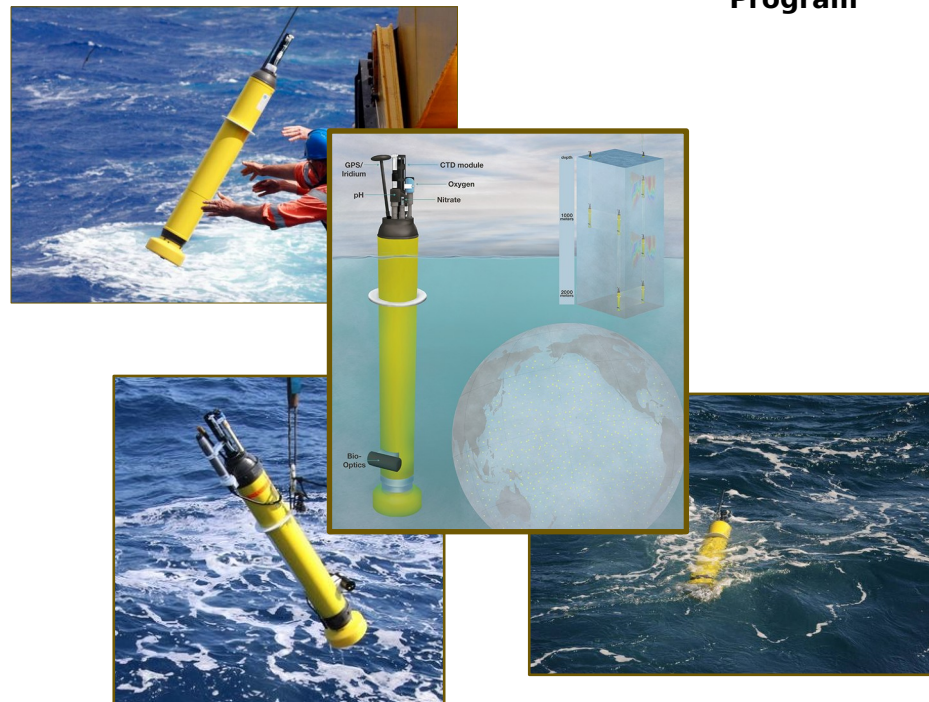


Biogeochemical Argo (BGC-Argo) is an extension of the global Argo program, designed to observe and monitor biogeochemical properties of the ocean using autonomous profiling floats.

<https://erddap.ifremer.fr/erddap/tabledap/ArgoFloats-synthetic-BGC.html>



Global ARGO Program



ARGO (BGC)

ERDDAP > tabledap

Tabledap lets you use the OPeNDAP constraint/selection protocol to request data subsets, graphs, and maps from tabular datasets (for example, buoy data). For a quick introduction, see this [video introduction to using tabledap](#). For details, see [ERDDAP's tabledap Documentation](#).

21 matching datasets, listed in alphabetical order. (Or, refine this search with [Advanced Search](#).)

Grid DAP Data	Sub-set	Table Data	Make A Graph	W M S	Source Data Files	Access-ible	Title	Summary	FGDC, ISO, Metadata	Back-ground Info	RSS	E-mail	Institution	Dataset ID
	set	data	graph		public		* The List of All Active Datasets in this ERDDAP *		F I M	background			Ifremer	allDatasets
		data	graph		public		2000-2014 climatology of the Subtropical Mode Waters and Permanent Pycnocline properties in the North Atlantic.		F I M	background			Ifremer/LOPS	OACP-Argo-NATL
		data	graph		public		Argo Float Measurements		F I M	background			Argo	ArgoFloats
	set	data	graph		public		Argo float synthetic vertical profiles : BGC data		F I M	background			Argo	ArgoFloats-synthetic-BGC

ERDDAP > tabledap > Data Access Form

Dataset Title: **Argo float synthetic vertical profiles : BGC data**

Institution: Argo (Dataset ID: ArgoFloats-synthetic-BGC)

Information: [Summary](#) | [License](#) | [FGDC](#) | [ISO 19115](#) | [Metadata](#) | [Background](#) | [Subset](#) | [Make a graph](#)

Variable ☐ Check All ☐ Uncheck All

- ☐ wmo_inst_type (Coded instrument type)
- ☐ data_type
- ☐ date_update (UTC)
- ☐ pi_name
- ☐ bbp470 (m-1)
- ☐ bbp470_qc
- ☐ bbp470_adjusted
- ☐ bbp470_adjusted_qc
- ☐ bbp470_adjusted_error
- ☐ bbp532 (m-1)
- ☐ bbp532_qc
- ☐ bbp532_adjusted
- ☐ bbp532_adjusted_qc
- ☐ bbp532_adjusted_error
- ☒ bbp700 (m-1)
- ☐ bbp700_qc (quality flag)
- ☐ bbp700_adjusted (m-1)
- ☐ bbp700_adjusted_qc (quality flag)
- ☐ bbp700_adjusted_error (m-1)

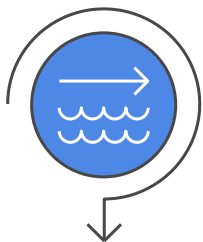
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>=	<=		
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>=	<=	" "	"0"
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<https://erddap.ifremer.fr/erddap/tabledap/ArgoFloats-synthetic-BGC.html>

CMEMS

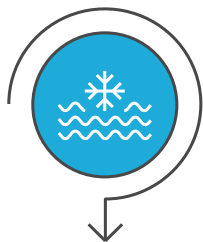
Copernicus Marine Environment Monitoring Service

Estimates of physical and biogeochemical variables and properties for the past, present and future ocean using mainly Observations



Blue ocean

Physical parameters like temperature, currents, and sea level.



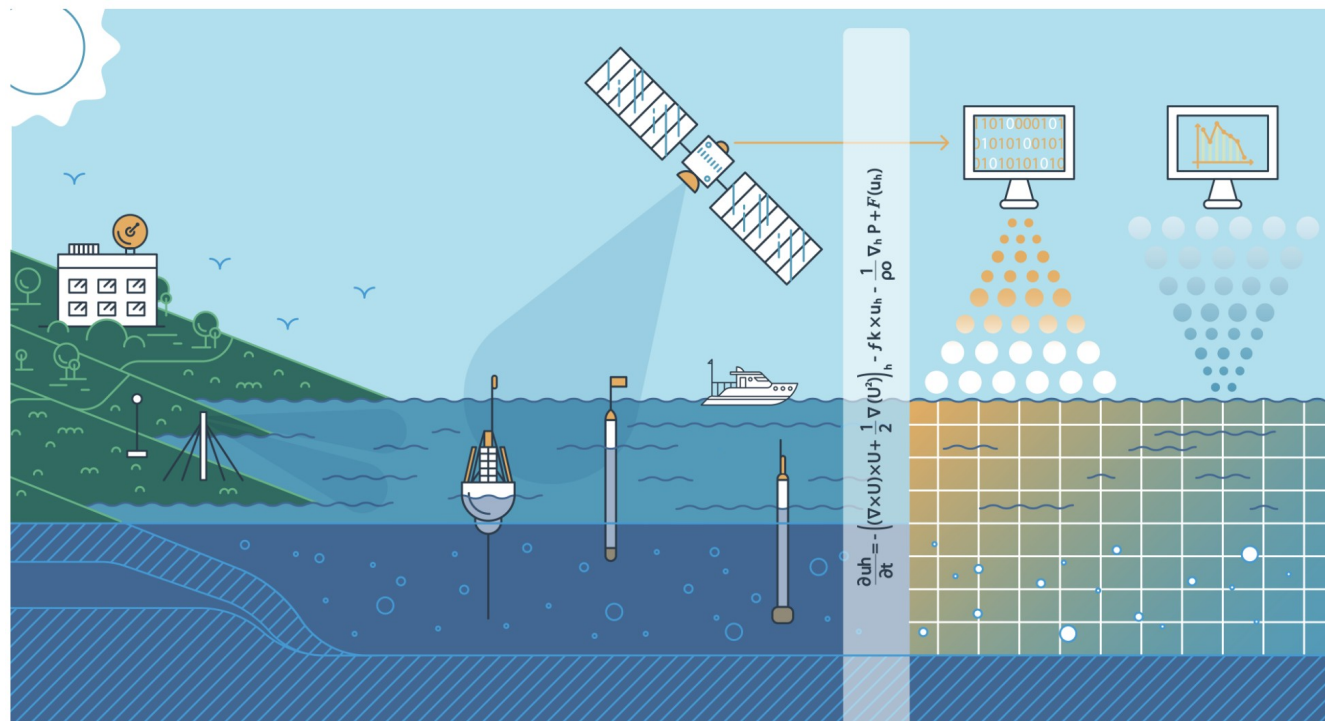
White ocean

Sea ice coverage and dynamics.



Green ocean

Biogeochemical aspects such as chlorophyll concentration, oxygen levels, and carbon content



Copernicus
Marine Service



Implemented by

**MERCATOR
OCEAN**
INTERNATIONAL

CMEMS

Copernicus Marine Environment Monitoring Service

- CMEMS:

<https://climate.copernicus.eu>

- Package dependencies:

```
> import copernicusmarine
```

- How to install toolbox:

<https://help.marine.copernicus.eu/en/articles/7970514-copernicus-marine-toolbox-installation>



- **Climate Reanalysis:** ECMWF (API-based data retrieval)
- **Climate In-situ:** ARGO (ERDDAP)
- **Climate Observation:** CMEMS (OpenDAP)

**EPOS
Data
Portal**

Application1:
epos_argo_cmems.ipynb
Application2: epos_ecmwf.ipynb



Thank you for your attention!

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www.epos-eu.org/on | social media [#EPOSon](#)